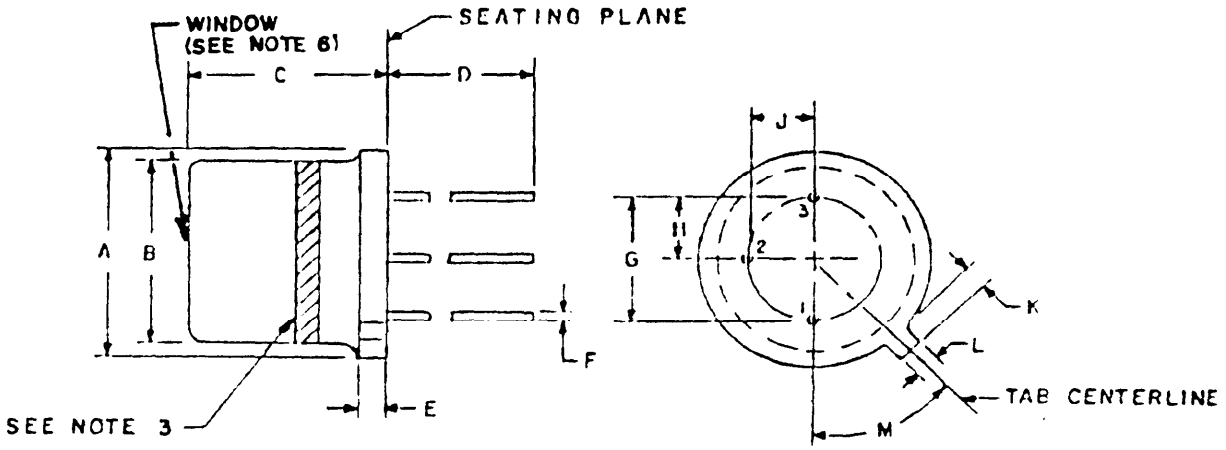
MILITARY SPECIFICATION SHEET

DETECTOR, APD, FIBER OPTIC, TYPE A AND B, CLASS 1, STYLE 3

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and and Standards (DODISS) specified in the solicitation: MIL-D-24620.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are for reference only.
3. A yellow band, a yellow dot, or a yellow mark shall be provided on the detector case to signify that this is a fiber optic device.
4. Pin 1, which is closest to the tab, shall be the anode conductor. Pin 2 is the cathode and pin 3 is internally connected to the case.
5. The lead conductors positional tolerance at the seating plane (T) shall be

⌖	ø .014 (0.36 mm) Ⓜ	T	A Ⓜ	L Ⓜ

where A and L are datums and T is a datum plane.

6. The optical window shall not extend beyond the face of the can more than 1 mil. There shall be no bubbles greater than 5 mils within the 40 mils diameter window.

FIGURE 1. ADP detector.

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Letter symbol	Inches		Millimeters	
	Min	Max	Min	Max
A	.209	.230	5.31	5.84
B	.183	.185	4.65	4.70
C	.123	.129	3.12	3.28
D	.500	---	12.70	---
E	---	.030	---	0.76
F	.016	.019	0.41	0.48
G	.100 T.P.		2.54 T.P.	
H	.050 T.P.		1.27 T.P.	
J	.050 T.P.		1.27 T.P.	
K	.033	.048	0.84	1.22
L	.039	.046	0.99	1.17
M	45° T.P.		45° T.P.	

FIGURE 1. PIN detector - Continued.

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degrees periodically during procedure. (NOTE:

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Operating temperature: -55°C to +125°C (M24620/3-01)

Storage temperature: -65°C to +150°C (M24620/3-01)

Terminal strength: Test condition E. The lead shall be bent three times with an 8-ounce lead bend for 15 seconds.

Lead conductors: Lead conductors shall be alloy-42 or Kovar (American wire guage 24) with a hot solder dip finish.

Mounting: Fiber optic alignment sleeves will be provided with MIL-C-83522 fiber optic connector receptacles.

Electrical and optical requirements: See table I.

Identification marking:

Part or Identifying Number (PIN): See table I.

Color band: A yellow band, yellow dot, or a yellow mark shall be included on the assembly case as shown on figure 1.

Marking: Marking size for detector type designation shall be 1 mm in height, and the manufacturer's designating symbol shall be identifiable without need for visual magnification.

Screening: Screening shall be conducted on 100 percent of all devices prior to groups A, B, and C inspections and shall be in accordance with the methods specified in table II.

Group A inspection: Group A inspection shall consist of the examinations and tests specified in table III.

Group B inspection: Group B inspection shall consist of the examinations and tests specified in table IV.

Group C inspection: Group C inspection shall consist of the examinations and tests specified in table V.

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TABLE I. Primary APD detector performance characteristics. 1/

Detector dash number M24620/3-	Sealing level type	Reverse voltage V_R (V) maximum 2/	Total capacitance C_T (pF) maximum 3/	Dark current I_D (nA) maximum 4/	Responsivity R (μA/μW) minimum 5/	Quantum efficiency (η %)	Rise and fall times (ns) t_r/t_f 6/	NEP W/ $\sqrt{Hz}$ 7/	Power dissipation mW maximum 8/
01	A	250	2	15	70	70 +10	.75	$3x10^{-15}$	100

1/ Unless otherwise specified, the ambient temperature shall be 30°C ±5°C. Also, the detectors covered by this specification sheet shall meet the dimensional requirements specified herein. However, for reference only, these detectors are similar to a TO-52 packaged device.

2/ V_R is the specified reverse bias voltage for the APD in volts. The reverse current, I_R shall be 10 μA.

3/ C_T is the APD junction capacitance in picofarads. Reverse voltage and frequency for capacitance shall be -5.0 volts and 1.0 MHz respectively.

4/ I_D is the APD dark current in nanoamperes. Reverse voltage for dark current shall be -5.0 volts with a 1 mΩ load.

5/ R is the APD responsivity in microamperes per microwatt. Reverse voltage of 5.0 volts, wavelength of 850 nm, and input power of 10 μW shall be employed. Values shown are for a detector connected to a 100/140 fiber possessing a .29 NA through an applicable MIL-C-83522 connector.

6/ Pulse rise and fall times shall be measured at a wavelength of 850 nm and a reverse bias voltage of 5.0 V.

7/ NEP is the APD noise-equivalent-power in watts per square root hertz with a modulation frequency f_m = 1.0 MHz.

8/ The total device power dissipation at an ambient temperature of 25°C. Above 25°C the power shall be derated at 0.5 mW/°C for M24620/3-01.

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TABLE II. Screening inspections.

Screening inspection (100%)	MIL-STD-750 method	Details
Internal visual (precap) inspection	2073	Ambient
High temperature (nonoperating) life (LTPD)	1032	24 hours minimum at T_A = +125°C for type A devices and +85°C for type B devices
Thermal shock (temperature cycling)	1051	Test condition A-1, low temperature = -46°C, high temperature = +125°C for type A and +85°C for type B.
Constant acceleration	2006	Y1 direction at 20,000 g's minimum at T_C = 25°C. The one minute hold time requirement shall not apply.
Hermetic seal	1071	
a. Fine leak rate 1/		Type A only, condition H.
b. Gross leak rate		Types A and B, condition C. Fluid temperature = +85°C
Electrical and optical measurements		
Dark current (I_D)	3061	Bias condition C, V_R = -5.0 V
Responsivity (R)		V_R = -5.0 V, λ = 850 nm P_I = 10 µW
Steady state reverse-bias burn-in	1038	Condition A, 80% maximum V_R for 48 hours at +100°C for type A and +70°C for type B

See footnotes at end of table.

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TABLE II. Screening inspections - Continued.

Screening inspection (100%)	MIL-STD-750 method	Conditions
Electrical and optical measurements 2/ ΔI_D ΔR		The permitted delta for I_D shall be 100% of 1 nA or 0.5 whichever is greater and R shall be 25% from initial values after burn-in
External visual and mechanical inspection 3/	2071	Ambient

1/ Sealing level type B sources may omit this test.
2/ If the quantity in the lot which fails the delta limits exceeds 10 percent the lot fails. If the percent defective exceeds 20 percent, the lot shall not be acceptable for any level. For those lots resubmitted for burn-in the PDA shall not exceed 3 percent.
3/ Visual and mechanical inspection shall be performed on all devices after marking has been completed.

TABLE III. Group A inspection.

Industry sublots	MIL-STD-750 method	Conditions	LTPD
Subgroup 1			
External visual and mechanical inspection	2071	Ambient	5
Subgroup 2		To be performed at 25°C.	
Dark current	3061	Bias condition C, V_R = -5.0 V	5
Responsivity		VR = -5.0 V, λ = 850 nm P_I = 10 µW	
Subgroup 3			
Breakdown voltage	4021	I_R = 10 µA	5
Capacitance	4001	V_R = -5.0 V F_M = 1 MHz	
Pulse response		V_R = -5.0 V λ = 850 nm	

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TABLE IV. Group B inspection.

Inspection	MIL-STD-750 method	Conditions	LTPD
Subgroup 1 1/			
Solderability 2/	2026		15
Resistance to solvents	1022		
Subgroup 2			
Thermal shock (temperature cycling)	1051		
Hermetic seal	1071		10
a. Fine leak		Type A, test condition G or H	
b. Gross leak		Types A and B test condition C or D. Fluid temperature = +85°C	
Electrical and optical measurements See group A, subgroup 2			
Subgroup 3 3/			
Steady state operation life (LTPD)	1027	340 hours, bias conditions as specified	5
Electrical and optical measurements See group A, subgroup 2			
Subgroup 4			
Decap internal visual design verification	2075		One device/0 failures for each lot
Bond strength	2037	The sample shall include a minimum of three devices	10 (C = 0)
Subgroup 5			
High temperature (nonoperating) life (LTPD)	1032		7
Electrical and optical measurements See group A, subgroup 2			

1/ Failed devices from the same inspection lot may be used for all subgroups when electrical or optical end-point measurements are not required.

2/ The LTPD for solderability test applies to the number of leads inspected except in no case shall less than three devices be used to provide the number of leads required.

3/ If a given inspection lot undergoing group B inspection has been selected to satisfy group C inspection requirements, the 340-hour life tests may be continued on test to 1,000 hours in order to satisfy the group C life test requirements. In such cases, either the 340-hour end-point measurements must be made as a basis for group B lot acceptance or the 1,000-hour end-point measurements shall be used for both groups B and C acceptance.

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TABLE V. Group C inspection.

Inspection subgroups	MIL-STD-750 method	Conditions 1/	LTPD
Subgroup 1			
Physical dimensions 2/	2066	See figure 1	15
Subgroup 2			
Thermal shock (glass strain)	1056	Test condition A	10
Terminal strength	2036	Test condition E	
Hermetic seal	1071		
a. Fine leak		Type A, test condition G or H.	
b. Gross leak		Type A or B, test condition C or D. Fluid temperature = +85°C	
Moisture resistance	1021		
Electrical and optical measurements See group A, subgroup 2			
Subgroup 3			
Shock	2016	Nonoperating 1500 g's, 0.5 ms, 5 blows in each orientation X1, Y1, and Z1	
Vibration	2056		
Constant acceleration	2006	One minute minimum in each orientation, X1, Y1, Z1 at 20,000 g's minimum T_C = 25°C	
Electrical and optical measurements See group A, subgroup 2			
Subgroup 4			
Salt atmosphere (corrosion)	1041		15
Subgroup 5 3/			
Steady state operation life (λ)	1025		λ = 10
Electrical and optical measurements See group A, subgroup 2			

1/ See requirement and test method paragraphs for complete exposure conditions.
2/ Electrical and optical reject devices from the same inspection lot may be used for all subgroups when electrical and optical end point measurements are not required.
3/ If a given inspection lot undergoing group B inspection has been selected to satisfy group C inspection requirements, the 340-hour life tests may be continued on test to 1,000 hours in order to satisfy the group C life test requirements. In such cases, either the 340-hour end-point measurements must be made as a basis for group B lot acceptance or the 1,000-hour end-point measurements shall be used for both groups B and C acceptance.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Navy - AS, EC
Air Force - 11, 17, 19, 80, 90, 99
DLA - ES
NASA - AS

User activities:
Navy - CG, MC
Air Force - 13, 14

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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